

23CS41SC –PROMPT ENGINEERING

Course Category:	Skill Enhancement Course	Credits:	2
Course Type:	Theory	Lecture-Tutorial-Practical:	0-1-2
Prerequisite:	Knowledge in Prompt Engineering	Sessional Evaluation: 30 Univ. Exam Evaluation: 70 Total Marks: 100	
Course Objectives:	Students undergoing this course are expected:		
	- Delves prompt engineering principles, strategies,. - Discuss a crucial aspect in shaping AI models' behaviour and performance - Understanding Prompt Engineering is a comprehensive course. - Understand the vision mechanisms of prompt engineering,		

Course Outcomes:	Upon successful completion of the course, the students will be able to:	
	CO1	Under standing the fundamentals and evolution of prompt engineering.
	CO2	Gaining the ability to craft effective closed-ended, open-ended, and role-based prompts
	CO3	Learning to probe and stress-test AI models for bias and robustness.
	CO4	Applying prompt optimization techniques and performance evaluation methods
	CO5	Mitigating bias and promoting ethical prompting practices in NLP/ML systems.
Course Content:	<u>Module-I</u> Introduction to Prompt Engineering: Lesson 1: Foundations of Prompt Engineering - Overview of prompt engineering and its significance in NLP and ML. - Historical context and evolution of prompt-based approaches	
	<u>Module -II</u> Types of Prompts and Their Applications: Lesson 2: Closed-Ended Prompts - Under standing and creating prompts for specific answers. - Applications in question- answering systems. Lesson 3: Open-Ended Prompts - Crafting prompts for creative responses. - Applications in language generation models	
	<u>Module -III</u> Strategies for Effective Prompting: Lesson 4: Probing Prompts - Designing prompts to reveal model biases.	

	○ Ethical considerations in using probing prompts. • Lesson 5: Adversarial Prompts ○ Creating prompts to stress-test models. ○ Enhancing robustness through adversarial prompting <u>Module -IV</u> Fine-Tuning and Optimizing with Prompts: • Lesson 6: Fine-Tuning Models with Prompts ○ Techniques for incorporating prompts during model training. ○ Balancing prompt influence and generalization. • Lesson 7: Optimizing Prompt Selection ○ Methods for selecting optimal prompts for specific tasks. ○ Customizing prompts based on model behavior <u>Module -V</u> Evaluation and Bias Mitigation • Lesson 8: Evaluating Prompt Performance ○ Metrics and methodologies for assessing model performance with prompts. ○ Interpreting and analyzing results. • Lesson 9: Bias Mitigation in Prompt Engineering ○ Strategies to identify and address biases introduced by prompts. ○ Ensuring fairness and inclusivity in prompt-based models <u>Module -VI</u> Real-World Applications and Case Studies: • Lesson 10: Case Studies in Prompt Engineering • Exploration of successful implementations and challenges in real-world scenarios. • Guest lectures from industry experts sharing their experiences.
Text Books & References Books:	TEXTBOOKS: 1. "Prompt Engineering in Action" – Danny D. Sullivan 2. "The Art of Prompt Engineering with Chat GPT: A Hands-On Guide" – Nathan Hunter REFERENCE BOOKS: 1. "Prompt Engineering in Practice" – Michael F. Lewis 2. "Mastering AI Prompt Engineering: The Ultimate Guide for Chat GPT Users" – Adriano Damiao 3. "Writing AI Prompts For Dummies" – Stephanie Diamond and Jeffrey Allan 4. "Prompt Engineering Guide" (Online Resource) – promptingguide.ai
E-Resources:	1. https://www.udemy.com/course/understandingpromptengineering/?couponCode=NVDINCTA35TRT